

# Ethan Xu

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## EDUCATION

University of California, Berkeley - Regents' & Chancellor's Scholar

Berkeley, CA

Computer Science B.A, 4.0 GPA

Expected May 2028

## TECHNICAL EXPERIENCE

Software Engineering Intern

Sept 2026 – Present

Coupage

- Building a computer vision and LLM pipeline against Coupage's production catalog, **converting raw product images and seller metadata into structured listings, attributes, SEO keywords, and quality scores**

Data Science Intern

Sept 2026 – Present

Climformatics

- Building a **meteorological forecasting model** based on GHI, DNI, DHI, temperature and wind speed data to generate predictions at an hourly resolution up to 14 days out, verifying against ground-truth solar data

Software Engineering Intern

Oct 2024 – Dec 2025

IPMD Inc. ASTAI Group

- Built Logistic Regression with output thresholds algorithms based on closing price fluctuations using Keras/Tensorflow that **outperformed S&P benchmarks by 13% over 10 years** across backtesting
- Developed and backtested Elliott Wave, candlestick pattern, and support/resistance trading strategies across **10 years of daily OHLCV SP 500 data**, achieving a **20% cumulative return** over the evaluation period.
- Implemented an **automated news ingestion pipeline** using BeautifulSoup to collect real-time data from Yahoo finance articles for sentiment analysis for multimodal model inputs

Lab Assistant, Dr. Christoph Eick

Jun 2025 – Sep 2025

Data Analysis and Intelligent Systems Lab, University of Houston

- Collected and mapped underwater elevation maps (NOAA) to **geofenced deep sea images** as training data
- Assisted in developing a CNN model to **automatically classify, label, and extract** front-facing horizon images from multiple online large-scale deep sea image datasets into model-ready specific image databases

Research Assistant, Dr. Melissa Baldo

Apr 2025 – Sep 2025

College of Business, Lamar University

- Researched sustainable short-sea freight shipping systems, evaluating economic and environmental benefits compared to standard trucking in the US in context with the **Jones Law** in a written report
- Completed **novel** case studies evaluating **GHG and PM emissions** relative to rail and trucking for real freight pipelines in Texas, designed and built all tables and images drafted into a paper for submission

**Fintrax (Project)** | Python, Pandas, BeautifulSoup API, Hugging Face API, Trainer API

- Developed a pretrained BERT based ML confidence model from Hugging Face based on **sentiment discrepancies** between prepared speeches and open Q&A sections in earning call statements from Motley Fool
- Processed and engineered financial datasets for training and **scraped real-time earnings calls** with BeautifulSoup to decide buy/sell orders based off of confidence score discrepancies

## PUBLICATIONS

**SENTINEL: Sentiment Evolution and Narrative Tracking in Extended LLM Interactions**

- Designed a framework to quantify intra-textual sentiment drift in long-context LLM interactions and agentic workflows, mitigating LLM emotional hallucination
- Published at the **NeurIPS MTI-LLM 2025 Workshop** [NeurIPS] [OpenReview]

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS

**Frameworks:** React, Next.js, Node.js, Flask, FastAPI, TensorFlow, Keras, PyTorch, Hugging Face Transformers

**Libraries:** pandas, NumPy, scikit-learn, Matplotlib, OpenCV, BeautifulSoup

**AI & Agentic Tools:** Claude Code, OpenAI Codex, Cursor, GitHub Copilot, OpenRouter, Fireworks AI

**Databases & Cloud:** PostgreSQL, MongoDB, Redis, Supabase, Firebase, AWS, Docker, Kubernetes

**Developer Tools:** Git, GitHub, VS Code, Jupyter, Google Colab, Figma